AWS CLOUD
(HOSPITAL MANAGEMENT SYSTEM)

AWS

AWS stands for Amazon Web Services. It is one of the most popular cloud computing platforms in the world. It provides computing, storage, networking, databases, security and many other services over the internet. Instead of buying expensive servers, I can use AWS resources whenever I need them. AWS follows a pay-as-you-use model, so I only pay for the services that my project actually uses. This makes it suitable for both small and large applications.

AWS Cloud

Cloud computing means storing data and running software on internet servers instead of our own computer. Users can access the application from anywhere using an internet connection. In a Hospital Management System, doctors, nurses, reception staff and administrators can use the same system from different departments. Cloud also makes software updates and maintenance easier because everything is managed in one place.

Amazon EC2

Amazon EC2 (Elastic Compute Cloud) provides virtual servers to run applications. I can install my Hospital Management System on an EC2 instance and make it available to users 24×7. EC2 allows me to increase CPU, memory or storage whenever more users start using the application. It is useful because hospitals may receive many requests at the same time and EC2 can handle this workload efficiently.

Use in Hospital Management System:
- Host the complete website.
- Handle multiple users simultaneously.
- Improve application performance.
- Upgrade server resources whenever required.

Amazon S3

Amazon S3 (Simple Storage Service) is an online object storage service. It is designed for storing large amounts of files securely. In my project I can save patient reports, prescriptions, laboratory reports, X-ray images, MRI scans, doctor profile photos, invoices and other important documents. Files stored in S3 are highly durable and can be accessed whenever needed.

Backup Files

Backup is very important because hospital data should never be lost. I can create automatic backups of patient reports and medical records in Amazon S3. Even if the local computer crashes or data is deleted accidentally, the backup copy will still be available. This improves reliability and reduces the risk of losing important medical information.

AWS Lambda

AWS Lambda is a serverless computing service that runs code automatically whenever a specific event occurs. There is no need to manage servers manually. I can use Lambda to send appointment confirmation emails, generate bills, send medicine reminder notifications and update patient records after specific actions. Since Lambda runs only when required, it also helps reduce server costs.

AWS Security

AWS provides different security services to protect applications and data. It supports user authentication, password protection, encryption, firewall rules, access control and automatic backups. These features help keep sensitive hospital information safe from unauthorized users.

Use in Hospital Management System:
- Doctors can update medical records.
- Reception staff can register patients.
- Patients can only view their own reports.
- All important data remains encrypted and protected.

AWS IAM (Identity and Access Management)

IAM controls who can access AWS resources and what actions they are are allowed to perform.

Admin: Full access to AWS services and Hospital Management System.
Doctor: Can view patient history, update treatment details and upload reports.
Reception Staff: Can register patients, book appointments and generate bills.
Patient: Can log in, check appointments, download reports and view prescriptions but cannot modify records.